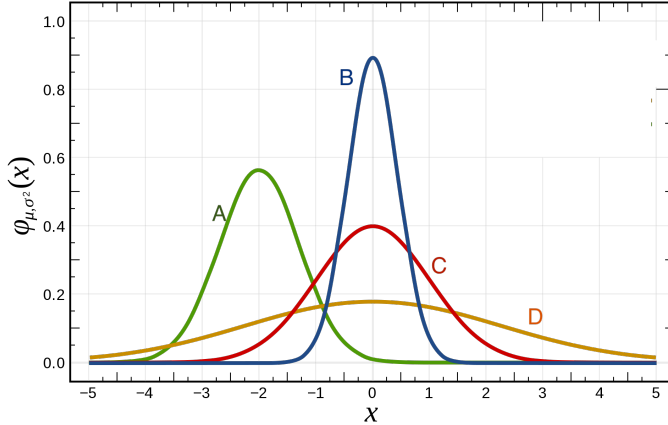


CSC311 Introduction to Machine Learning Week 10 Exercises

When submitting problems, please **clearly indicate which question you are submitting**. Use the full question number shown below.

Training

W10-T1. Which of the following are true about the parameters (μ, σ^2) for each of these four univariate Gaussian distributions?



- (a) Distribution D has the largest variance.
- (b) Distributions A, B, D have the same mean.
- (c) Distributions A has a higher variance than distribution C.

W10-T2. Suppose we would like to model the distribution of data $\mathcal{D} = \{x^{(1)}, x^{(2)}, \dots, x^{(N)}\}$ using a univariate Gaussian distribution:

$$\mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

- (a) Show that the log likelihood of the data can be written

$$\ell(\mu, \sigma^2) = -\frac{N}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^N (x^{(i)} - \mu)^2$$

- (b) Show that the maximum likelihood estimate of μ is

$$\mu = \frac{1}{N} \sum_{i=1}^N x^{(i)}$$

- (c) Show that the maximum likelihood estimate of σ^2 is

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x^{(i)} - \mu)^2$$

W10-T3. Consider the following data set $\mathcal{D} = \{3, 5, 8, 4, 6, 2, 7\}$ consisting of 7 observations. Assume that these data points were drawn independently and identically distributed (i.i.d.) from a Gaussian distribution with unknown mean μ and unknown variance σ^2 .

The probability density function (PDF) of a univariate Gaussian distribution with mean μ and variance σ^2 is given by:

$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

For a dataset $\mathcal{D} = \{x_1, x_2, \dots, x_N\}$ of N i.i.d. observations from a Gaussian distribution, the maximum likelihood estimates (MLE) for the mean and variance are given by:

$$\hat{\mu}_{MLE} = \frac{1}{N} \sum_{i=1}^N x_i$$

$$\hat{\sigma}_{MLE}^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu}_{MLE})^2$$

- Compute the MLE estimate for the mean for the given data set.
- Compute the MLE estimate for the variance for the given data set.
- Suppose we add a new data point 10 to our dataset. Without recalculating everything, explain how this would affect our estimates of $\hat{\mu}_{MLE}$ and $\hat{\sigma}_{MLE}^2$.

W10-T4. Explain how the contours of each of these pairs of bivariate Gaussian distributions differ

- Same mean across both distributions, but $\Sigma_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $\Sigma_2 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$
- Same mean across both distributions, but $\Sigma_1 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ and $\Sigma_2 = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}^{-1} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$
- Same covariance across both distributions, but $\mu = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $\mu_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

W10-T5. Suppose we would like to build a Gaussian Discriminate Analysis model of the data

$$\mathcal{D} = \{(x^{(1)}, t^{(1)}), (x^{(2)}, t^{(2)}), \dots, (x^{(N)}, t^{(N)})\},$$

where $x^{(i)} \in \mathbb{R}$, and $t^{(i)} \in \{0, 1\}$. Recall that the PDF of a univariate Gaussian is

$$p(x; \mu, \sigma^2) = \mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- List all the parameters of this GDA model. Define each parameter.
- Suppose that we have the following data set $\mathcal{D} = \{(5, 0), (10, 1), (12, 1)\}$. Write down the log-likelihood of the data, $\log p(\mathcal{D})$, plugging in all probability density functions and all possible numbers.

Validation

W10-V1. In this question, we will show that for a binary Gaussian distribution, the contours of the probability density function forms an ellipse. A multivariate Gaussian distribution has the following probability density function

$$\mathcal{N}(\mathbf{x}) = \frac{1}{(2\pi)^{D/2} |\Sigma|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})\right]$$

In this question, we will assume that $\mathbf{x} \in \mathbb{R}^2$, so $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Thus, $\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$, and $\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12}^2 \\ \sigma_{12}^2 & \sigma_2^2 \end{bmatrix}$,

- (a) Recall that in a contour plot, a contour of a function f consists of points in $\mathbf{x} \in \mathbb{R}^2$ for which $f(\mathbf{x})$ is a constant. Show that a contour of the function $\mathcal{N}(\mathbf{x})$ has

$$(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) = C$$

for some constant $C \in \mathbb{R}$.

- (b) Show that this above set can be written in the form $c_1 x_1^2 + c_2 x_2^2 + c_3 x_1 x_2 + c_4 x_1 + c_5 x_2 + c_6 = 0$.
 $\boldsymbol{\Sigma}^{-1} = \begin{bmatrix} s_1 & s_2 \\ s_2 & s_3 \end{bmatrix}$ (Note: why must this matrix be symmetric?)
- (c) Explain why the above is sufficient to show that the contours are ellipses.
- (d) Show that if x_1 and x_2 are uncorrelated (i.e., $\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$), the contours of the Gaussian distribution consists of axis-aligned contours. That is, we can write the contours in the form $\mathcal{E}(x_1, x_2) = c_1(x_1 - d_1)^2 + c_2(x_2 - d_2)^2 + \mathcal{E}_0$

W10-V2. You are given that in Gaussian Discriminant Analysis (GDA) with two classes (class 0 and class 1), we model:

- $P(c = k) = \phi_k$ (class prior)
 - $P(x|c = k) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}_k|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu_k)^T \boldsymbol{\Sigma}_k^{-1} (x - \mu_k)\right)$ (class-conditional densities)
- (a) Show that if $\boldsymbol{\Sigma}_0 = \boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}$ (i.e., both classes share the same covariance matrix), then the decision boundary between the two classes is linear.
- (b) Show that if $\boldsymbol{\Sigma}_0 = \boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}$, the GDA model can be written $p(c|\mathbf{x}) = \frac{1}{1 + \exp(-\mathbf{w}^T \mathbf{x} - b)}$ for some choice of $\mathbf{w}$ and b . Show this by writing out the values of $\mathbf{w}$ and b in terms of the parameters of the GDA model.
- (c) How does this compare to the logistic regression model? When should we prefer GDA to logistic regression and vice-versa?

Generalization

W10-G1. You are given that in Gaussian Discriminant Analysis (GDA) with two classes (class 0 and class 1), we model:

- $P(c = k) = \phi_k$ (class prior)
- $P(x|c = k) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}_k|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu_k)^T \boldsymbol{\Sigma}_k^{-1} (x - \mu_k)\right)$ (class-conditional densities)

Show that when $\boldsymbol{\Sigma}_0 \neq \boldsymbol{\Sigma}_1$ (different covariance matrices for each class), the decision boundary is a conic section (for 2D data) or a quadric surface (for higher dimensions).